

Analysis First-Year Exam, May 2018, Part 2

Answer *FOUR* questions in detail. State carefully any results used without proof.

1. This problem has two parts.
 - (i) Let $g_n \rightarrow g$ and $h_n \rightarrow h$ uniformly on X . Prove that if g and h are bounded, then $g_n h_n \rightarrow gh$ uniformly on X .
 - (ii) Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous. By considering the square-root or otherwise, prove that if f is non-negative then there exists a sequence of non-negative polynomials converging uniformly to f on $[a, b]$.
2. Let $\mathcal{F}$ be an equicontinuous family of real-valued functions on the connected space X . By considering the set of points at which the family $\mathcal{F}$ is bounded, prove that if $\mathcal{F}$ is bounded at one point of X then it is bounded at all points of X .
3. $(f_n : n \geq 0)$ is a sequence of measurable real-valued functions. Prove that each of the following sets is measurable:
 - (a) the set A comprising all points ω for which the sequence of values $f_n(\omega)$ is eventually positive;
 - (b) the set B comprising all points ω for which the sequence of values $f_n(\omega)$ changes sign infinitely often.
4. **Warning: Suspected error.** The source clearly says “Lebesgue-measurable functions,” but measurability and uniform convergence alone do not ensure the stated integrability conclusion, even when the measure is finite. A missing integrability hypothesis may have been intended, but the correction is not uniquely determined, so the source wording has been retained.

$(\Omega, \mathcal{F}, \mu)$ is a measure space on which $(f_n : n \geq 0)$ is a sequence of Lebesgue-measurable functions converging uniformly to f . Prove that if $\mu(\Omega) < \infty$ then it follows that f is integrable and $\int_{\Omega} f_n d\mu \rightarrow \int_{\Omega} f d\mu$. Show by example that if $\mu(\Omega) = \infty$ then the same conclusion may not follow.
5. Let $(f_n : n \geq 0)$ be a uniformly-bounded sequence of Riemann-integrable functions on $[a, b]$. If $f_n \rightarrow 0$ pointwise, does it follow that $\int_a^b f_n \rightarrow 0$? Proof or counterexample required.